Crop Recommendation Analysis Using Unsupervised Machine Learning

Submitted By:

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- **Subject:** Machine Learning

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1. Introduction

Agriculture plays a vital role in ensuring food security and economic development. Farmers often face challenges in selecting suitable crops because soil nutrients and environmental conditions vary significantly across regions. Data-driven techniques can help analyze these conditions and discover patterns that support agricultural decision-making.

Unsupervised Machine Learning is useful when datasets do not contain predefined labels for training. Clustering algorithms can group similar observations based on their characteristics, allowing us to identify hidden patterns in agricultural data.

In this project, clustering techniques are applied to soil and environmental features to group similar agricultural conditions. These groups provide insights into crop-growing environments and demonstrate how unsupervised learning can support agricultural analysis.

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2. Problem Statement

Farmers require accurate information about soil nutrients and environmental conditions to make informed crop planning decisions. However, identifying similar agricultural conditions manually is difficult because of the large amount of data involved.

The objective of this project is to use clustering algorithms to group similar agricultural conditions based on soil nutrients and weather-related parameters. These clusters help identify hidden structures within the dataset and provide insights into crop recommendation.

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3. Objectives

The objectives of this project are:

- To analyze agricultural soil and environmental data.
- To perform data preprocessing and exploratory data analysis.
- To apply multiple clustering algorithms.
- To compare clustering performance using evaluation metrics.
- To visualize clusters using Principal Component Analysis (PCA).
- To interpret the discovered clusters.

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4. Dataset Description

The **Crop Recommendation Dataset** was obtained from Kaggle.

The dataset contains **2,200 records** with **8 attributes**.

Feature	Description
N	Nitrogen content in soil
P	Phosphorus content
K	Potassium content
temperature	Temperature (°C)
humidity	Relative Humidity (%)
ph	Soil pH
rainfall	Rainfall (mm)
label	Crop name

Since this project uses **unsupervised learning**, the **label column was excluded during model training** and used only for interpreting the clusters.

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5. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked for missing values.
- Checked for duplicate records.
- Verified data types.
- Removed the crop label before clustering.
- Standardized numerical features using StandardScaler.

Feature scaling was necessary because clustering algorithms rely on distance calculations, and all features should contribute equally.

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6. Exploratory Data Analysis (EDA)

Several visualization techniques were used to understand the dataset:

- Histograms
- Boxplots
- Correlation Heatmap
- Pair Plot
- Distribution Plot

Observations

- The dataset contains no missing values.
- Soil nutrients have different distributions.
- Rainfall exhibits greater variation than other variables.
- Most features have weak to moderate correlations.
- A few outliers were observed but retained because they represent real agricultural conditions.

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7. Methodology

The project followed the following workflow:

Dataset
↓
Data Cleaning
↓
EDA
↓
Feature Scaling
↓
K-Means Clustering
↓
DBSCAN
↓
Hierarchical Clustering
↓
Evaluation
↓
Visualization
↓
Conclusion

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8. Algorithms Used

K-Means Clustering

K-Means partitions the dataset into K clusters by minimizing the distance between data points and cluster centroids.

Advantages:

- Simple
- Fast
- Efficient for numerical data

DBSCAN

DBSCAN groups data based on density and identifies outliers automatically.

Advantages:

- Detects noise.
- Works with irregular cluster shapes.

Hierarchical Clustering

Hierarchical Clustering builds a hierarchy of clusters using distance-based merging.

Advantages:

- Produces a dendrogram.
- Does not require iterative centroid updates.

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9. Model Evaluation

The clustering models were evaluated using:

- **Silhouette Score**
- **Davies–Bouldin Index**
- **Calinski–Harabasz Score**

K-Means Results

```
••• Silhouette Score : 0.25526916389002857
    Davies-Bouldin Index : 1.386621974857969
    Calinski-Harabasz Score : 658.6969498661996
```

DBSCAN Results

```
•• Silhouette Score: 0.22899920335558085
   Davies-Bouldin Index: 0.9286065441296132
   Calinski-Harabasz Score: 288.4481221320182
```

Hierarchial Results

```
••• Silhouette Score: 0.2592304693088373
    Davies-Bouldin Index: 1.426909735115703
    Calinski-Harabasz Score: 629.472501911232
```

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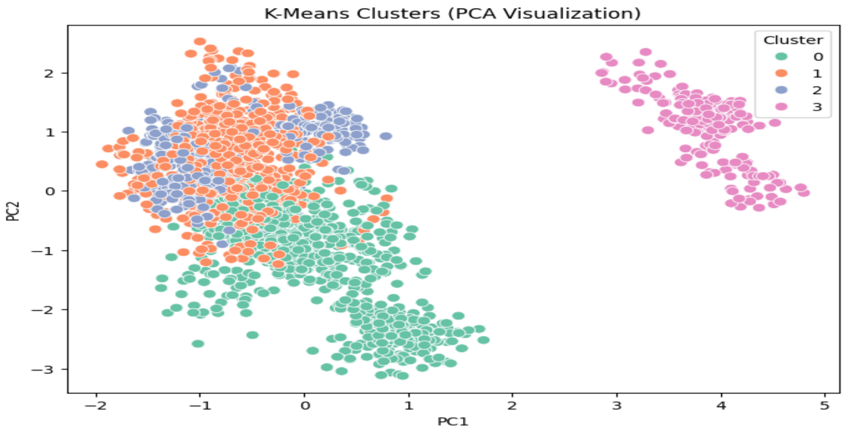
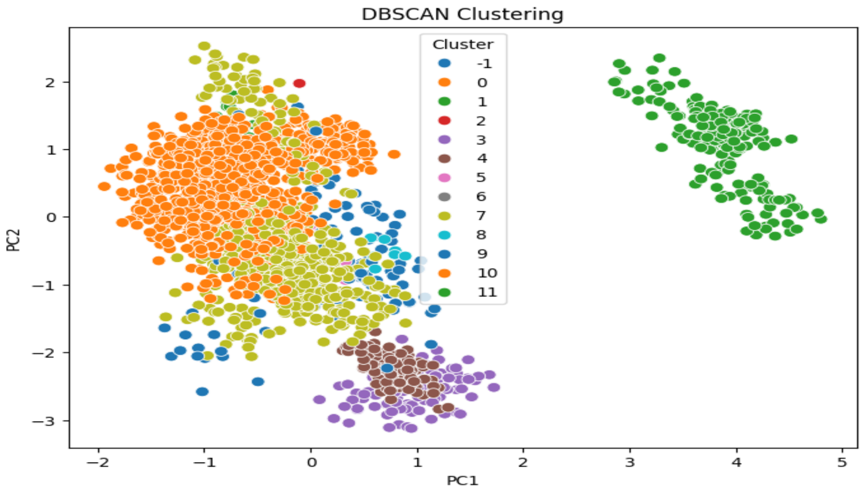
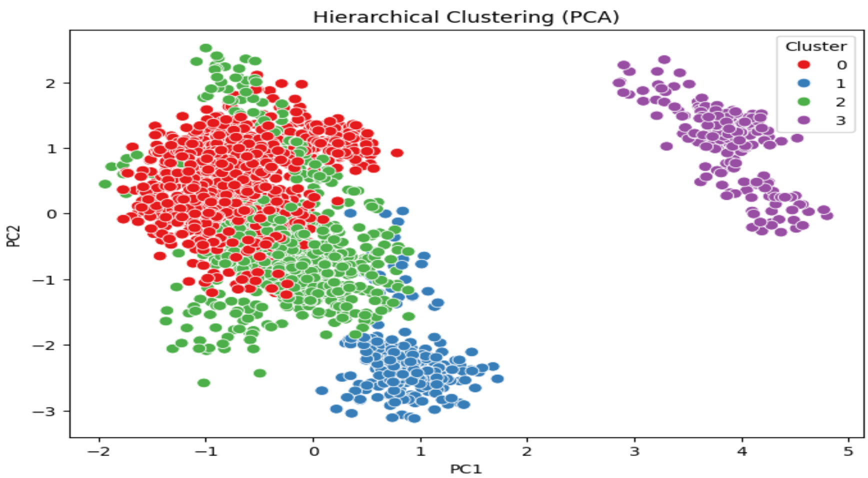
10. Cluster Visualization

Principal Component Analysis (PCA) was used to reduce the seven numerical features to two principal components. This made it easier to visualize the clusters in a two-dimensional plot.

The PCA visualization showed that one cluster was clearly separated, while some overlap existed among the remaining clusters due to similarities in agricultural conditions.







11. Results and Discussion

The clustering algorithms successfully grouped agricultural conditions with similar soil nutrients and weather characteristics.

Key findings:

- K-Means produced meaningful clusters with reasonable separation.
- PCA confirmed the presence of distinct cluster structures.
- Similar agricultural conditions were grouped together.
- The clusters can assist in understanding crop-growing environments.

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12. Applications

This project can be applied in:

- Precision Agriculture
- Smart Farming
- Soil Health Analysis
- Crop Planning
- Agricultural Decision Support Systems

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13. Conclusion

This project demonstrated the use of unsupervised machine learning techniques for analyzing agricultural data. After preprocessing and exploratory data analysis, clustering algorithms were applied to group similar agricultural conditions.

Among the implemented methods, K-Means produced meaningful clusters with satisfactory evaluation metrics. The PCA visualization further confirmed the effectiveness of the clustering process. The discovered clusters provide useful insights into agricultural conditions and can support informed crop planning.

Future work may include incorporating additional environmental factors, experimenting with advanced clustering methods, or integrating supervised learning models for crop prediction.

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14. Future Scope

Future improvements could include:

- Using larger agricultural datasets.
- Incorporating satellite imagery.
- Integrating weather forecasts.
- Developing a web-based crop advisory system.
- Comparing additional clustering algorithms.

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15. References

1. Kaggle – Crop Recommendation Dataset.
2. Scikit-learn Documentation.
3. Pandas Documentation.
4. NumPy Documentation.
5. Matplotlib Documentation.
6. Seaborn Documentation.

Appendices

- **Dataset preview**

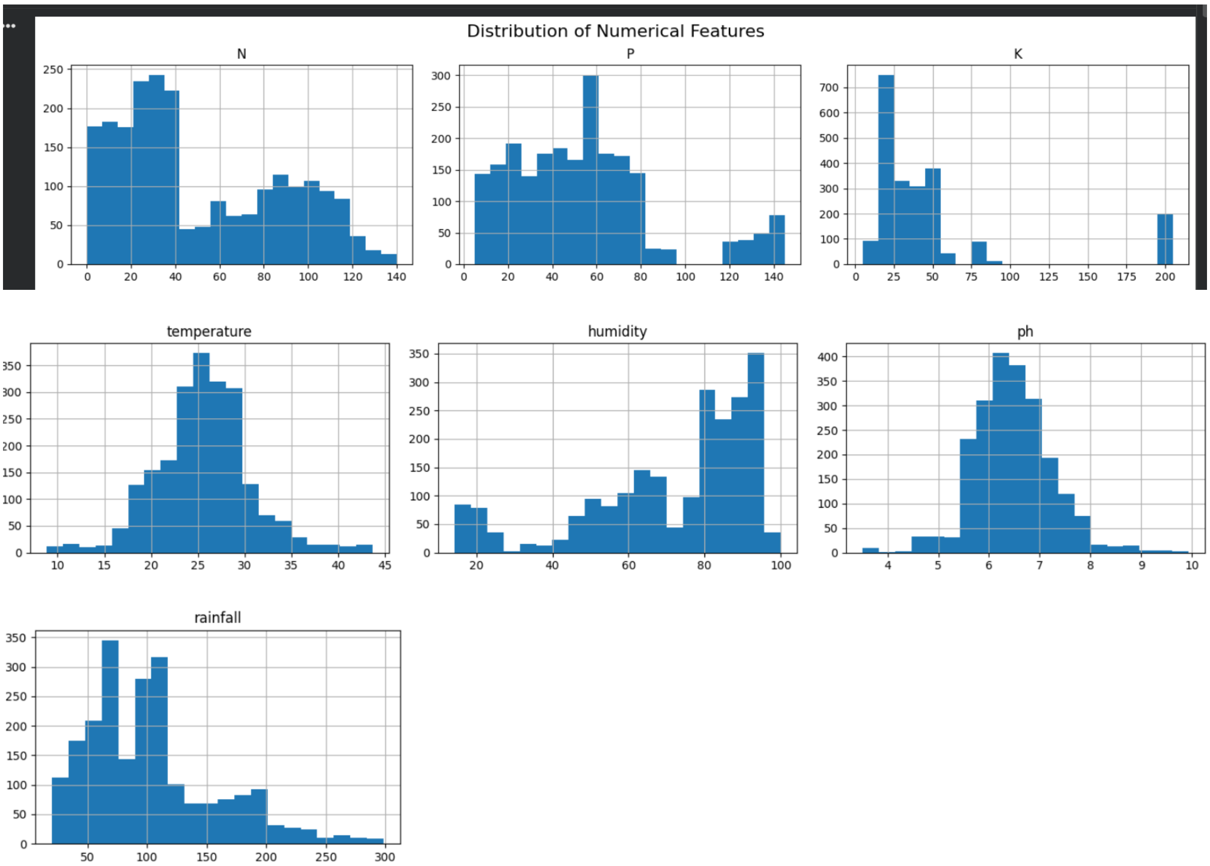
```
==================================================
Dataset Shape
==================================================
(2200, 8)


==================================================
Dataset Information
==================================================
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2200 entries, 0 to 2199
Data columns (total 8 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   N            2200 non-null   int64
 1   P            2200 non-null   int64
 2   K            2200 non-null   int64
 3   temperature  2200 non-null   float64
 4   humidity     2200 non-null   float64
 5   ph           2200 non-null   float64
 6   rainfall     2200 non-null   float64
 7   label        2200 non-null   object
dtypes: float64(4), int64(3), object(1)
memory usage: 137.6+ KB
```

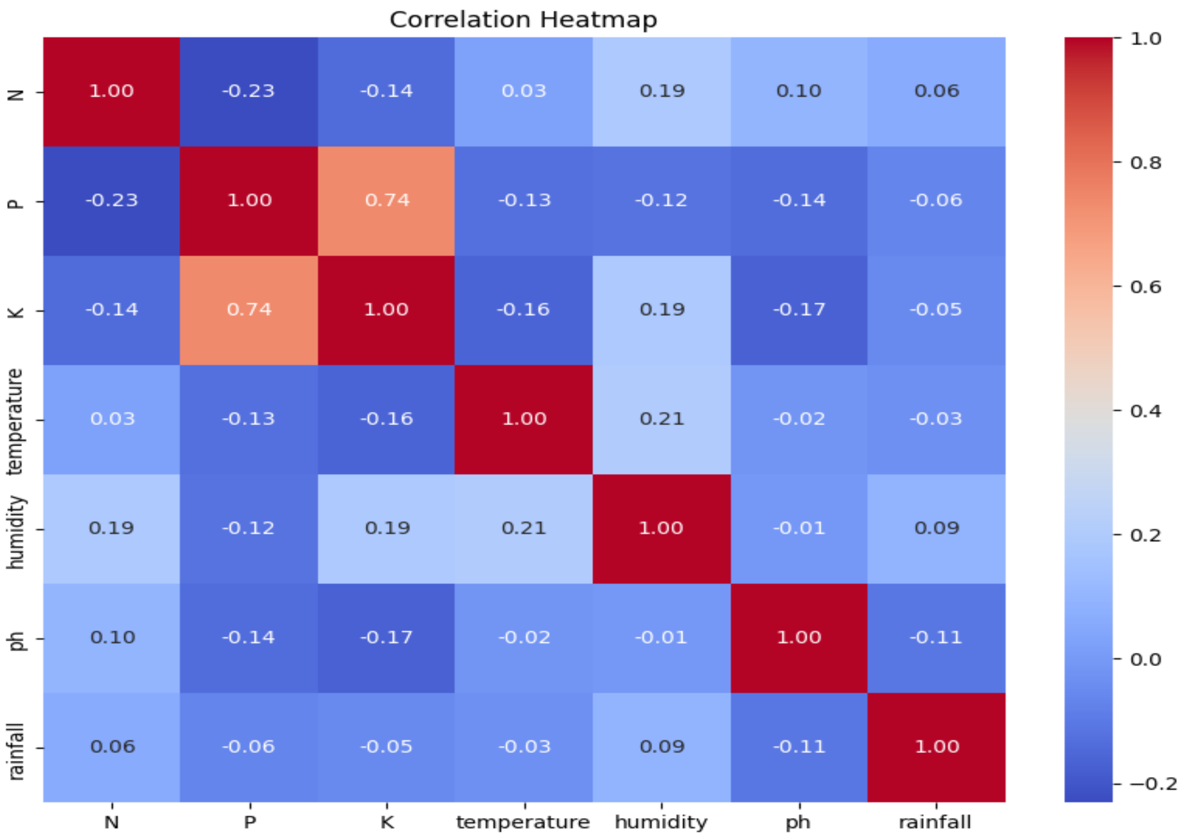
```
==================================================
Statistical Summary
==================================================
```

	N	P	K	temperature	humidity	ph	rainfall
count	2200.000000	2200.000000	2200.000000	2200.000000	2200.000000	2200.000000	2200.000000
mean	50.551818	53.362727	48.149091	25.616244	71.481779	6.469480	103.463655
std	36.917334	32.985883	50.647931	5.063749	22.263812	0.773938	54.958389
min	0.000000	5.000000	5.000000	8.825675	14.258040	3.504752	20.211267
25%	21.000000	28.000000	20.000000	22.769375	60.261953	5.971693	64.551686
50%	37.000000	51.000000	32.000000	25.598693	80.473146	6.425045	94.867624
75%	84.250000	68.000000	49.000000	28.561654	89.948771	6.923643	124.267508
max	140.000000	145.000000	205.000000	43.675493	99.981876	9.935091	298.560117

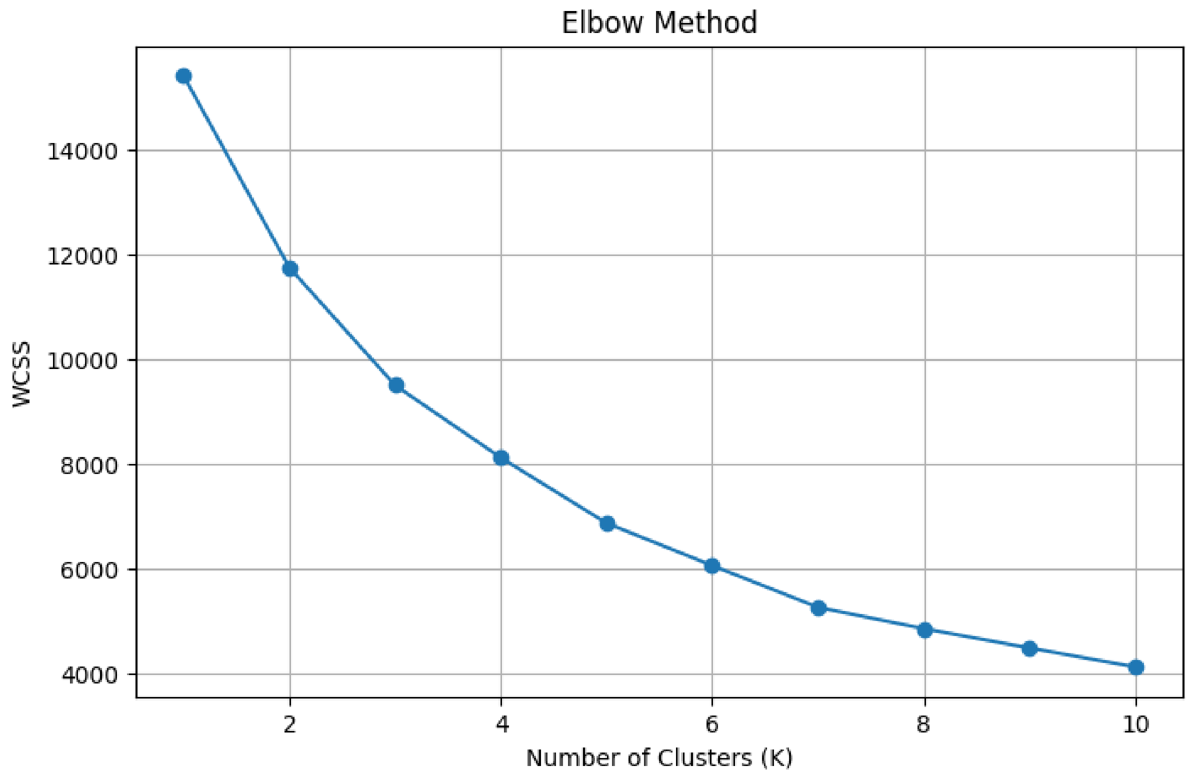
- **Histograms**



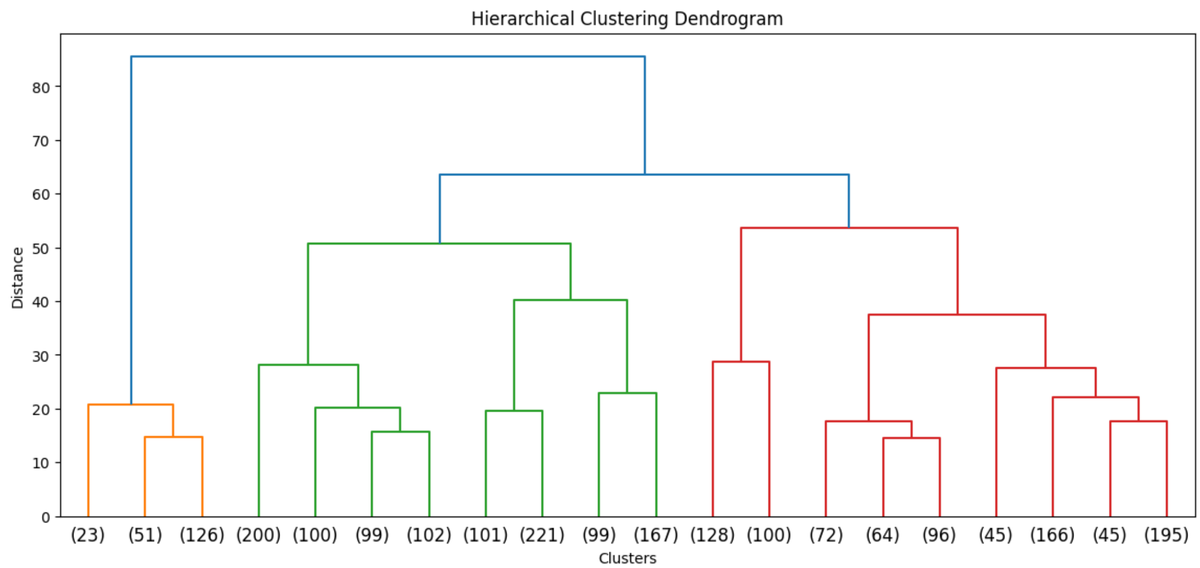
- **Correlation Heatmap**



- **Elbow Method graph**



- **Dendrogram**



QUESTION-ANSWER:

1. Semantic vs Instance vs Panoptic Segmentation

Semantic Segmentation	Instance Segmentation	Panoptic Segmentation
Assigns a class label to every pixel in an image.	Identifies and separates each individual object of the same class.	Combines semantic and instance segmentation by labeling every pixel and distinguishing different object instances.
Objects of the same class are treated as one group.	Each object has its own unique identity.	Provides both class information and object identity.
Does not distinguish between multiple objects of the same type.	Separates multiple objects of the same class.	Gives a complete understanding of the scene.
Example: All crop plants are labeled as "Crop."	Example: Each individual crop plant is identified separately.	Example: Every crop plant is identified separately, while the soil and background are also labeled.

Example

Suppose an image contains **three mango trees**.

- **Semantic Segmentation:** All three trees are labeled as **Tree**.
- **Instance Segmentation:** Each tree is labeled separately (Tree 1, Tree 2, Tree 3).
- **Panoptic Segmentation:** Every pixel is classified, and each tree is identified individually while the background (soil, sky) is also labeled.

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2. What is Fuzzy Logic, and how is it different from Boolean Logic?

Fuzzy Logic

Fuzzy Logic is a method of reasoning that allows values to lie **between 0 and 1**, representing degrees of truth instead of absolute true or false values. It is useful for handling uncertainty and imprecise information.

Example

If the temperature is **30°C**, Fuzzy Logic may interpret it as:

- Hot = 80%
- Warm = 20%

Thus, the temperature can belong to multiple categories simultaneously.

Boolean Logic

Boolean Logic works with only two possible values:

- True (1)
- False (0)

Example

If the condition is:

Temperature > 30°C

- 31°C → True
- 29°C → False

There is no intermediate value.

Difference Between Fuzzy Logic and Boolean Logic

Fuzzy Logic	**Boolean Logic**
Allows partial truth values between 0 and 1.	Only two values: True or False.
Handles uncertainty effectively.	Cannot represent uncertainty.
Suitable for real-world decision-making.	Suitable for exact logical conditions.
Used in intelligent systems, robotics, and control systems.	Used in digital circuits and computer programming.

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3. Customer Dataset Question

Question:

Suppose you are given a customer dataset containing:

- Age
- Annual Income
- Spending Score

but no labels.

(a) Is this a supervised or unsupervised learning problem?

Answer:

This is an **unsupervised learning problem** because the dataset does not contain any target labels or predefined categories. The goal is to discover hidden patterns or groups within the data using clustering algorithms.

(b) Which clustering algorithm would you choose first and why?

Answer:

I would choose **K-Means Clustering** as the first algorithm because:

- It is simple and easy to implement.
- It is computationally efficient.
- It works well with numerical data.
- It performs effectively when clusters are compact and roughly spherical.
- It is widely used for customer segmentation problems.

(c) What assumptions does that algorithm make about the data?

Answer:

K-Means Clustering makes the following assumptions:

1. The number of clusters (**K**) is known in advance.
2. The clusters are approximately spherical in shape.
3. The clusters have similar sizes and densities.
4. Numerical features are used for distance calculations.
5. The data is scaled appropriately before clustering.
6. Each data point belongs to only one cluster.